

WUDAOLIANG, China

WMO: 529080

Lat: 35.2156N

Lon: 93.0806E

Elev: 4613

StdP: 56.87

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.4	(c) -23.6	(d) -36.8	(e) 0.2	(f) -16.4	(g) -35.0	(h) 0.2	(i) -15.1	(j) 18.1	(k) -8.1	(l) 16.8	(m) -7.2	(n) 2.5	(o) 250	(p) 0.941

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.6	(c) 15.7	(d) 6.0	(e) 14.0	(f) 5.4	(g) 12.5	(h) 4.8	(i) 8.0	(j) 13.0	(k) 7.2	(l) 11.6	(m) 6.3	(n) 10.5	(o) 4.5	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.1	10.5	10.2	5.3	9.9	9.2	4.5	9.4	8.4	38.2	13.2	35.7	11.8	33.2	10.5	10.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	12.8	10.8	-28.5	18.5	1.5	1.8	-29.6	19.7	-30.5	20.8	-31.4	21.7	-32.5	23.0
			-29.0	9.3	1.5	0.9	-30.1	10.0	-30.9	10.5	-31.7	11.1	-32.8	11.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.8	-15.0	-12.8	-9.2	-4.2	0.5	4.1	7.0	6.7	3.2	-3.5	-9.8	-13.5
	DBStd	8.34	3.26	3.67	3.57	3.45	2.56	2.58	2.42	2.39	2.63	3.43	2.96	3.12
	HDD10.0	5049	774	637	596	424	293	176	96	106	205	420	593	729
	HDD18.3	8083	1032	871	854	674	552	426	350	361	455	678	843	987
	CDD10.0	8	0	0	0	0	0	1	4	3	0	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.3	5.7	5.9	5.3	4.6	4.1	3.7	3.4	3.2	3.2	3.5	4.6	5.1
Precipitation	PrecAvg	264	1	2	3	7	21	46	70	64	42	7	1	1
	PrecMax	407	4	5	9	19	48	110	133	129	84	37	9	6
	PrecMin	136	0	0	0	1	3	8	25	16	3	0	0	0
	PrecStd	48	1	1	2	4	11	20	29	26	19	7	2	1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.1	0.9	5.5	9.0	12.8	15.6	18.6	18.4	13.6	9.9	2.0	-0.2
		MCWB	-7.6	-6.6	-4.0	-1.8	0.7	5.5	7.5	6.2	5.4	0.1	-5.6	-7.4
	2%	DB	-4.6	-1.6	2.4	6.9	10.3	13.1	16.0	15.8	11.4	6.2	-0.2	-3.1
		MCWB	-10.0	-8.1	-5.5	-2.8	0.0	4.3	6.4	6.0	4.4	-1.3	-6.8	-9.1
	5%	DB	-6.4	-3.5	0.4	5.1	8.3	11.3	14.2	13.9	9.8	3.9	-1.7	-5.1
		MCWB	-11.1	-9.2	-6.6	-3.5	-0.8	3.7	5.7	5.6	3.7	-2.4	-7.7	-10.4
	10%	DB	-8.2	-5.3	-1.3	3.3	6.4	9.5	12.6	12.2	8.2	2.0	-3.2	-6.9
		MCWB	-12.3	-10.2	-7.5	-4.3	-1.4	3.1	5.2	5.2	2.8	-3.4	-8.5	-11.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.8	-6.4	-3.7	-0.6	2.7	8.1	9.5	8.8	7.1	1.5	-5.2	-7.2
		MCDB	-1.9	0.2	4.6	6.1	7.5	12.1	15.2	13.6	11.3	6.2	1.3	-0.5
	2%	WB	-9.8	-7.9	-5.2	-2.0	1.3	6.4	8.3	7.9	5.9	-0.2	-6.5	-9.0
		MCDB	-5.0	-2.1	1.7	4.8	6.8	10.5	13.4	13.1	10.3	4.3	-0.9	-3.4
	5%	WB	-10.9	-9.0	-6.3	-3.0	0.4	5.2	7.3	7.2	4.9	-1.4	-7.4	-10.2
		MCDB	-6.5	-4.0	-0.1	3.8	5.7	9.2	11.9	11.7	8.4	2.6	-2.3	-5.3
	10%	WB	-12.1	-10.1	-7.3	-3.8	-0.5	4.1	6.5	6.3	3.9	-2.6	-8.3	-11.3
		MCDB	-8.3	-5.6	-1.8	2.3	4.7	7.8	10.6	10.4	6.8	1.1	-3.6	-7.1
Mean Daily Temperature Range	5% DB	MDBR	12.7	13.1	14.0	12.6	10.4	9.4	9.6	9.8	9.1	9.9	12.6	12.9
		MCDBR	14.1	14.7	15.8	14.8	13.8	12.1	12.5	13.0	11.2	11.5	13.7	14.5
		MCWBR	10.1	9.9	9.9	7.5	5.7	5.2	4.9	5.0	5.1	5.4	8.5	10.1
	5% WB	MCDBR	13.8	14.0	15.2	13.2	10.7	9.8	9.3	9.7	9.3	9.9	12.9	14.2
		MCWBR	9.9	9.6	9.6	6.9	5.3	5.0	4.5	4.6	5.0	5.1	8.3	9.9
Clear-Sky Solar Irradiance	taub		0.174	0.198	0.246	0.286	0.307	0.302	0.295	0.287	0.258	0.224	0.184	0.169
	taud		2.607	2.539	2.387	2.269	2.271	2.399	2.488	2.510	2.561	2.581	2.611	2.649
	Ebn at Noon		1090	1080	1036	998	974	974	979	986	1010	1032	1067	1083
	Edh at Noon		70	86	111	133	135	119	108	103	93	82	70	63
All-Sky Solar Radiation	RadAvg		3.29	4.09	5.23	6.21	6.56	6.40	6.51	6.00	5.23	4.51	3.71	3.20
	RadStd		0.22	0.27	0.28	0.27	0.31	0.36	0.45	0.34	0.36	0.32	0.17	0.13

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.50	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		+0.56	+1.12	N/A	N/A	+0.36	+0.26	-244	-254	N/A	N/A

Nomenclature: See separate page